Laboratoire 3 : Pont en H – Groupe B18

1. Alimenter le moteur en redressement synchrone

To work with “redressement synchrone” we will want to manipulate all transistors (T1, T2, T3, T4) given in the configuration seen in Figure 1.

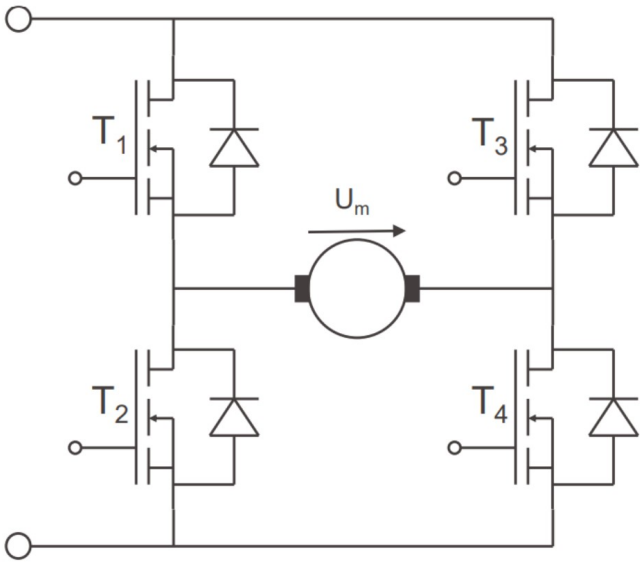


Figure 1: Schema for H-bridge (from class notes)

In redressement synchrone, there are two PWM signals: one controls T1 and T2, and the other controls T3 and T4. Both signals alternate between one transistor and the other, with dead time in between to prevent short-circuiting. The initial default state is one where the two PWM signals are synchronized – as seen in Figure 2, the motor experiences no voltage and as such does not move. The control triggers between the two top transistors and the two bottom transistors at the same time, creating zero current.

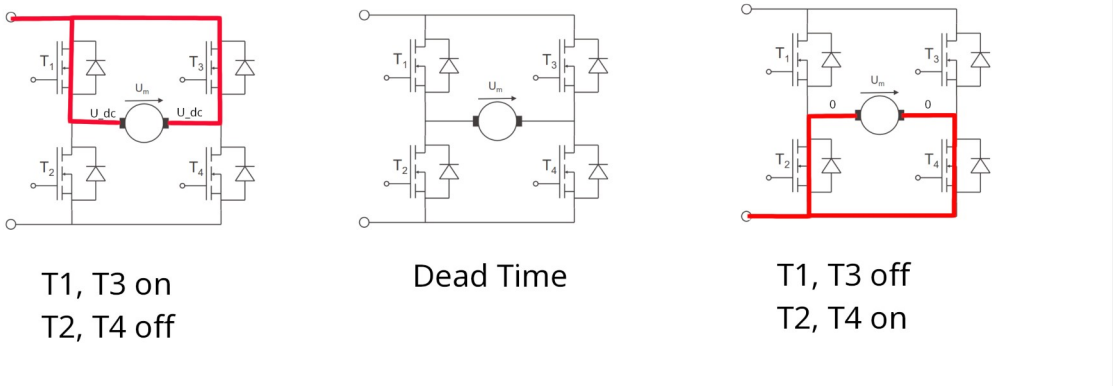


Figure 2: Demonstration when both branches have equal "taux"

Motor movement results from varying the duty cycle of the two PWM signals so that one diagonal (T1&T4 or T2&T3) is ON longer than the other, creating an overall current flow through the motor. For this, we define:

$$taux = (P/4096) * PWM_{period}$$

Knowing that (P) is the potentiometer reading, then scaled to match the range of PWM magnitudes. By defining the values for the Compare Registers as

$$CCR1 = taux$$
$$CCR2 = PWM_{period} - taux$$

We create two extremities at the minimum and maximum values of the potentiometer, with 0 corresponding to T1 and T4 held always ON, and 4096 corresponding to T2 and T3 always ON. Ignoring the discharging of the bootstrap capacitor, these two states correspond to maximum average voltage in the two directions. Figure 3&4 shows the oscilloscope reading for 25% potentiometer, for low and high motor loads respectively. Green and blue are the current and voltage through the motor, while yellow and red are the voltages across T1 and T4:

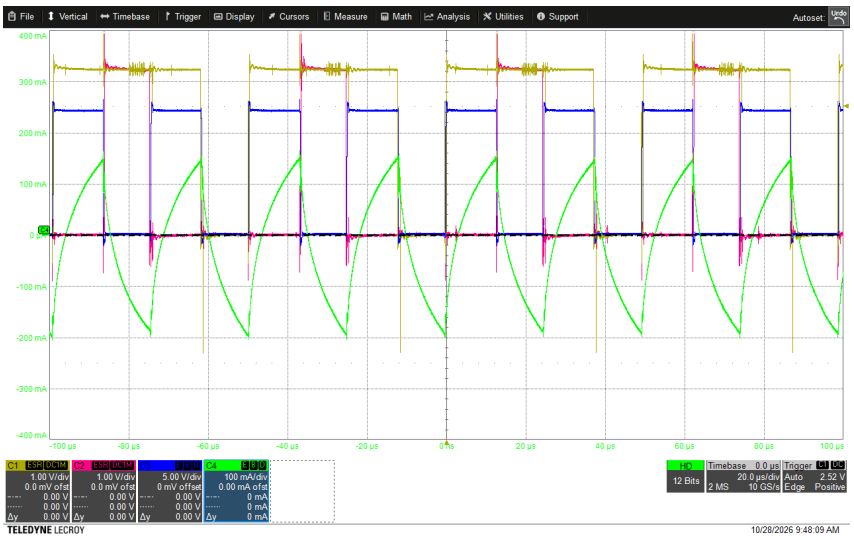


Figure 3: Redressement Synchrone with no external load

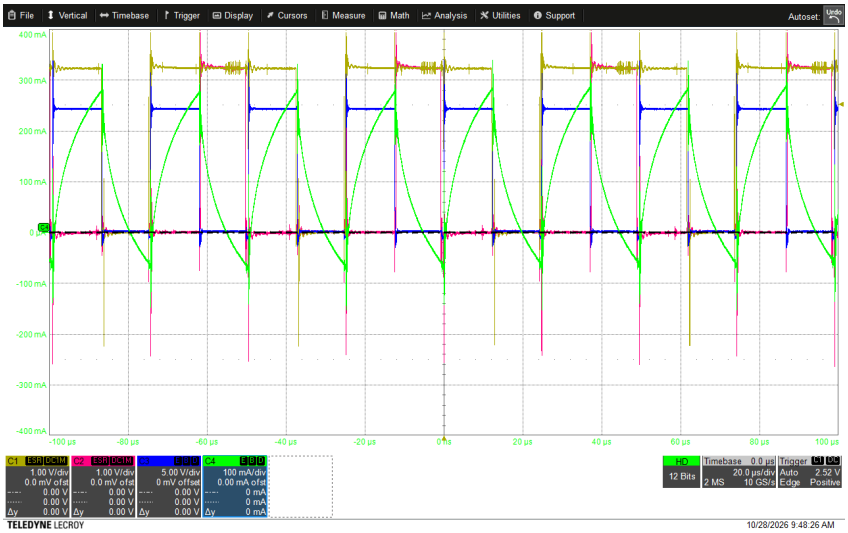


Figure 4: Redressement Synchrone with external load (potentiometer set to 25)

A brief analysis of what is happening approximately above is shown below, Note that the colours are meant to represent which transistor is on, under the respective branch's PWM signal:

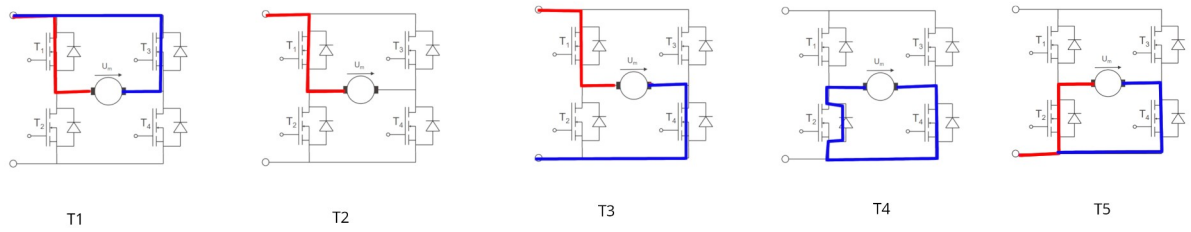


Figure 5: Diagram demonstrating when there is a asymetric difference bewteen Left and Right Branch Duty Cycle

- **Step 1:** Initially both top transistors are open, voltage difference across motor is 0.
- **Step 2:** Transistor 3 (top-right) closes as it has a smaller duty cycle, 25%. Dead time as Transistor 4 hasn't turned on yet
- **Step 3:** The transistor 4 (bottom right) turns on, and stays on. Now the motor has a voltage differential of + U_{DC}
- **Step 4:** The transistor 1 turns off, and momentarily the current has to flow through the freewheeling diode on T2 during the dead-time within the left branch.
- **Step 5:** The transistor 2 turns on, and the current can flow smoothly through the low-resistance transistor instead of the diode.

By controlling the duty cycle of the branches asymmetrically, we were able to replicate and ensure that there exists an average potential difference across the motor, and that this will then result in the motor being able to operate. We observe that this particular form of control has two current cycles per PWM cycle, which can lead to some energy losses. We can also observe the benefit with "redressement synchrone" is that for a low load on the motor, the current will keep flowing, as it passes through the transistor and not the diode, gently going under 0A and then remounting, whereas with a diode it would shut off completely, meaning the motor speed feels smooth and not jerky at low loads.

2. Alimenter le moteur en PWM double

In "*PWM Double*" we exert control over both diagonal transistors, switching between either T1 and T4 set to ON or T2, T3 set to ON. This is done by repeatedly applying a mask to the Capture/Compare Register, which then allows or blocks the PWM signal from reaching each transistor. In the "*PWM Double*" situation, we turn both diagonal transistors off / on, and when they are off, they have no free-path except through the freewheeling opposite diagonal diodes, to maintain the current direction flow of the motor, back into main power supply. As the current now flows from Ground (0V) to power supply (U_{DC}), creating a swing from + U_{DC} to -U_{DC}. This results in harsher spikes of current, as the motor experiences a larger range of voltages within the same time period as other control methods.

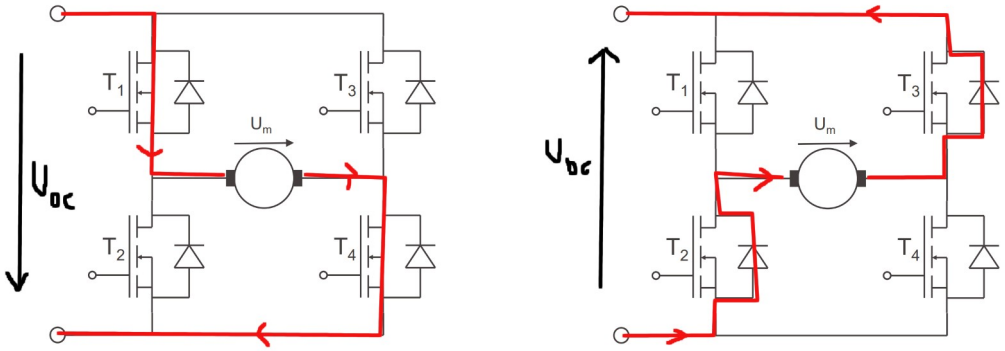


Figure 6: PWM Double turned on and off

In the figure above we see the cycle, from when the transistors are on, to when they are turned off (the motor still demands the current, and the inductance requires it, and the only available path is the one shown in the flow to the right. Figure 7 does a good job of showing the dramatic voltage change at each PWM cycle – there is a brief spike downward to roughly -12V which quickly becomes positive again.

An interesting thing we observed was when we put the potentiometer all the way to the left (i.e. longer duty cycle) the motor would stutter temporarily and then go back to rotating. The reason for this is the bootstrap did not have enough time to recharge. This was due to the rapid voltage spikes before settling, meaning the freewheeling diodes conduct but for a very small duration. The capacitor charges during this period, and as the time it has to recharge is so small, it doesn't have enough charge to last through higher duty cycles. Turning to the other side doesn't affect because you are keeping a low duty cycle, hence no problem.

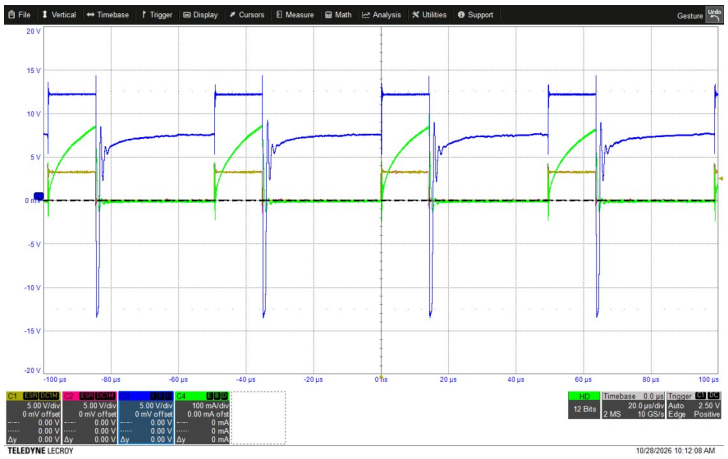
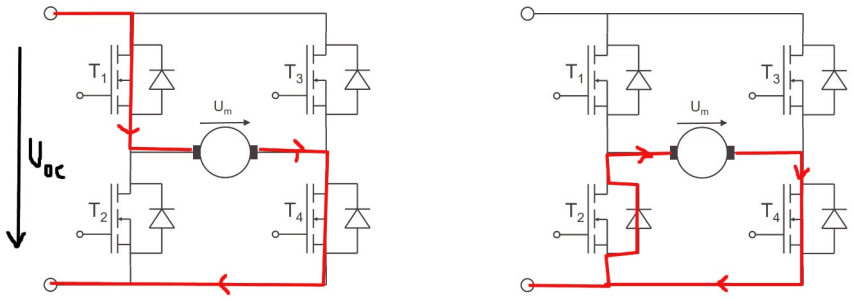


Figure 7: PWM Double with Potentiometer on 75%

3. PWM Simple

In "*PWM Simple*", the extent of control is reduced to only the top transistor, while the diagonally opposite bottom transistor stays on always. Interestingly to note, while the issue of the bootstrap capacitor become fully discharged is still possible in this configuration, this only occurs for much smaller PWM values than with PWM double. Here, the voltage across the motor only oscillates between 0 and -12V, which is much less aggressive than in PWM double, giving the bootstrap capacitor more time to recharge. The reasoning can be explained by the fact that Voltage drops to 0 not -12V each PWM cycle, as current instead of flowing through the opposite diagonals diodes, flows through the loop at the bottom, as shown below.



This means voltage drops to 0V and not -12V during the same cycle, resulting in a smoother, longer climb, less violent current spikes, and consequently less energy loss. This extra time gives the capacitor longer to charge up, and hence it can work at higher duty cycles, which we observed by putting a high duty cycle and motor did not stutter. This can be seen in Figure 8 – the rate of decrease of current on PWM falling edge is much more gentle than in Figure 7, allowing for a greater amount of charge to accumulate each PWM cycle.

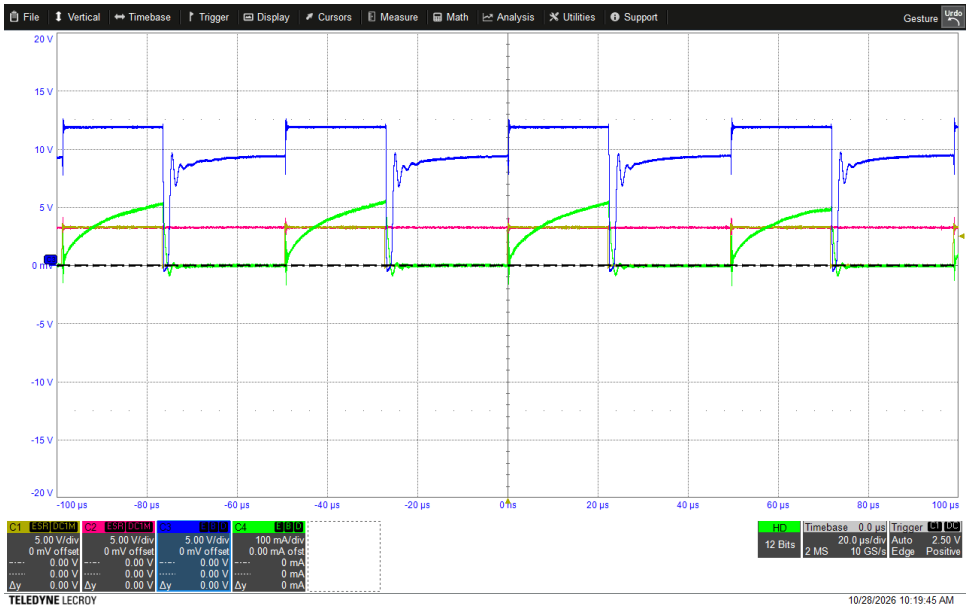


Figure 8: PWM-Simple, with potentiometer at the center

4. <u>Tourner moteur dans le deux sens (en PWM Simple)</u>

Inherently in the PWM Simple case, we only alternate between Source Voltage and 0V, meaning the motor only ever turns in one direction. We want to extend control to both directions. The idea here is to use all transistors, and adapt the potentiometer so its 50% value reading is the point of lowest absolute voltage across the motor. Going further from this middle point should increase the magnitude of voltage across the motor, flowing in one direction or another, as seen in Figure 9. We need to adapt our scaling and usage of the potentiometer value, to switch between the branches that are active and also their duty cycle, following the control plan laid out in Figure 10.

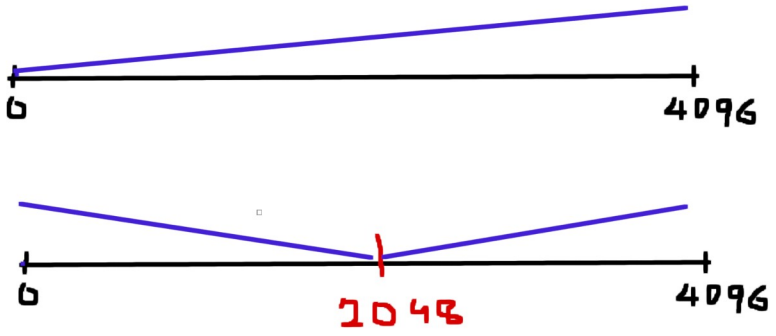


Figure 9: Demonstration on how to convert Potentiometer values to turn motor both sides in PWM Simple

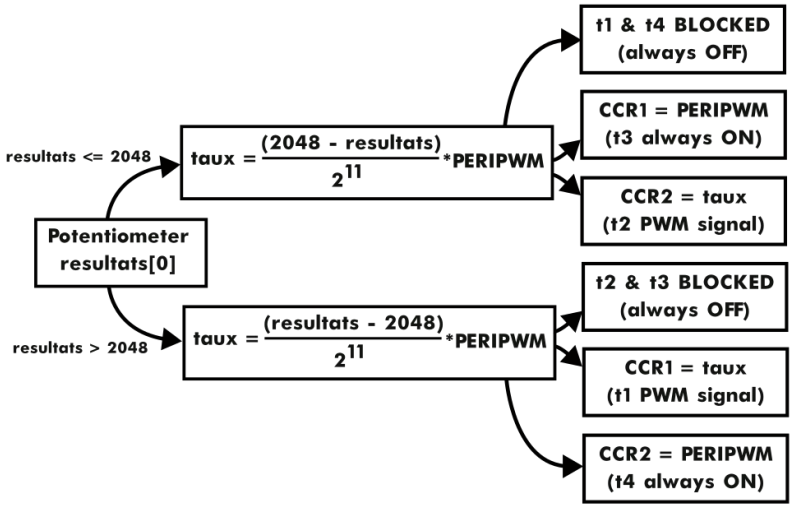


Figure 10: Flowchart determining how potentiometer value controls each transistor

This type of control is essentially switching between two configurations of simple PWM, according to the direction the voltage should flow in. The similarities between Figure 11 and Figure 8 can be seen, with the same drop from 12V to 0V when the PWM turns off the upper transistor. Figure 12 shows the second configuration of this control method, when the potentiometer passes the middle point – the voltage and current behaviour is fundamentally the same, except reflected in the x-axis as negative values.

During the programming and building of this control method, a curious problem was initially observed – everything would work as expected when switching the power supply, but after a couple of seconds the output behaved unexpectedly – during the upper transistor's ON state, the voltage would curve up without quite reaching 12V, then stabilise at a much lower value. For this reason, the current through the motor was also lower than expected and the motor's decrease in speed could be heard by ear. We identified the issue to be a mix-up in the control of the two active transistors: the upper transistor was set to be always on while the lower transistor was modulated with PWM, the opposite of the intended control. This meant that the bootstrap capacitor quickly discharged completely and could no longer provide the full voltage needed to keep the upper transistor always ON.

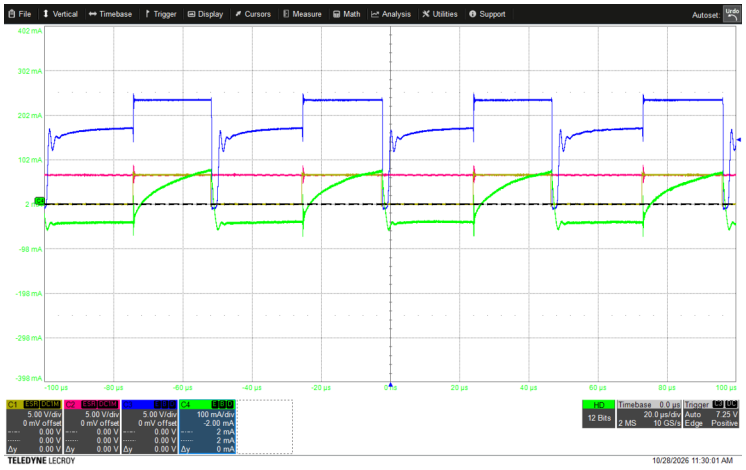


Figure 11: Potentiometer at 50%, T1 and T4 active

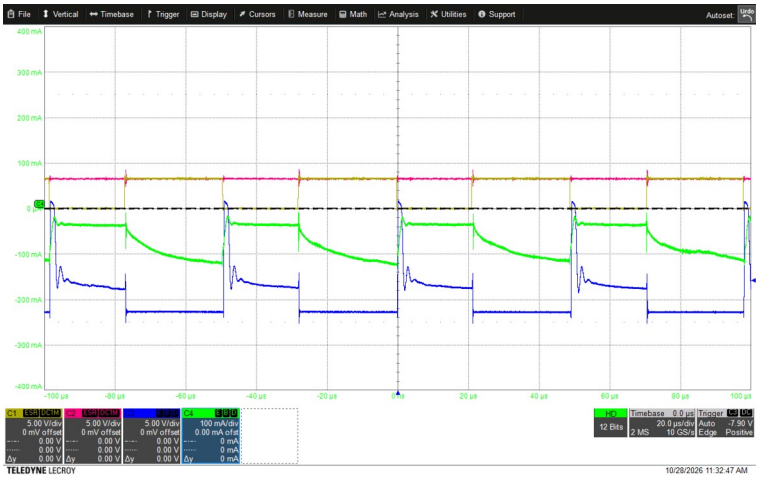


Figure 12: Potentiometer at 75%, T2 and T3 active

Figure 13 and Figure 14 show the two Simple PWM control methods both close to the ‘zero’ point (50% on the potentiometer). Because the ON signal from the PWM is very brief, we see the voltage across the motor quickly go up to the full 12V then back down to zero almost instantly, a sign that there is little back emf from the slow-spinning motor. The current, however, takes much longer to decay to a constant value than for higher PWM signals (e.g. in Figure 12). This is also a result of the lower back emf, as it means the current is forced down less quickly and the current is more strongly characterised by the inductance in the coils.

The measurements of voltage and current were taken close to but not at 50% on the potentiometer, because setting the potentiometer too close to the dividing point created a chaotic, difficult to follow signal. At some point, the potentiometer is close enough to the middle that random noise in the results[0] reading constantly brings the value above and below the middle signal, which the controller tries to keep up with by constantly switching which transistors are ON and OFF. This creates a wildly oscillating voltage signal through the motor, possibly creating large energy losses. An improvement for the program to avoid this would be creating a buffer area around 50% before switching to another side that would account for any measurement noise.

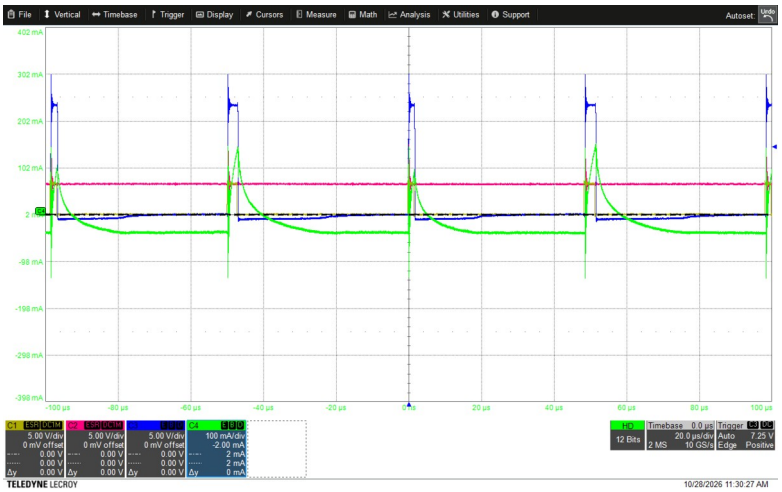


Figure 13: When Potentiometer is at 50% viewed by T1 and T4

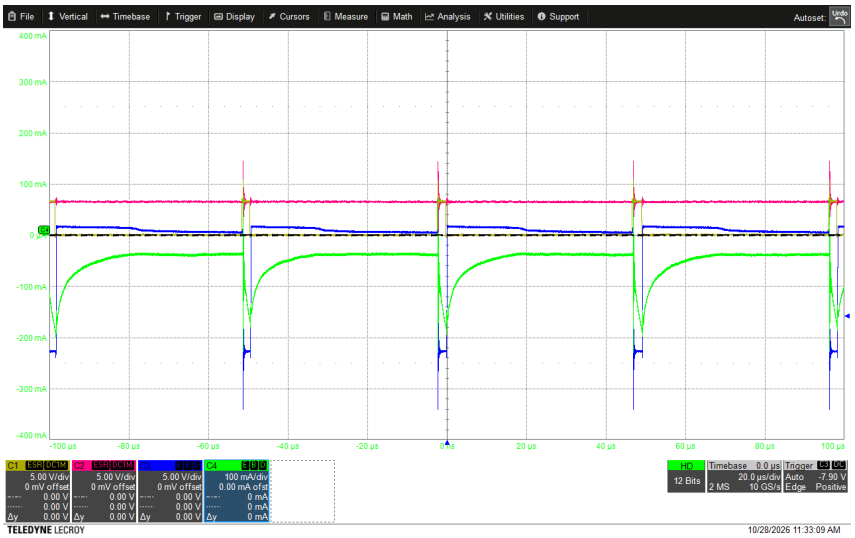


Figure 14: When Potentiometer is at 50% viewed by T2 and T3